Nicholas R. Beaton

Contact

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Employment

The University of Melbourne, Parkville, Victoria, Australia

School of Mathematics and Statistics

Senior Lecturer in Mathematical Physics — **September 2023 – present**
Lecturer in Mathematical Physics — **January 2022 – August 2023**
Research Fellow — **May 2021 – January 2022**
ARC DECRA Research Fellow — **January 2017 – April 2021**

The University of Saskatchewan, Saskatoon, Saskatchewan, Canada

Department of Mathematics and Statistics

PIMS Postdoctoral Fellow — **December 2014 – December 2016**

The University of Melbourne, Parkville, VIC, Australia

ARC Centre of Excellence for Mathematics and Statistics of Complex Systems (MASCOS)
Department of Mathematics and Statistics

Research Assistant and Casual Lecturer — **January – December 2014**

Laboratoire d'Informatique de Paris Nord (LIPN)

Université Paris-Nord (Paris 13)
Villetaneuse, France

Postdoctoral Researcher (ANR Project MAGNUM) — **November 2012 – October 2013**

Education

The University of Melbourne, Parkville, VIC, Australia

Doctor of Philosophy — **2009 – 2012**

The University of Queensland, St. Lucia, QLD, Australia

BSc (Hons) – Mathematics — **2005 - 2008**

Teaching Experience

* *indicates coordinating as well as lecturing.*

The University of Melbourne

MAST20009 Vector Calculus — **Semester 1, 2026**
*MAST30012 Discrete Mathematics** — **Semester 2, 2025**
*MAST20009 Vector Calculus** — **Semester 1, 2025**
*MAST90031 Enumerative Combinatorics** — **Semester 2, 2024**
*MAST90067 Advanced Methods: Transforms** — **Semester 1, 2024**
MAST10006 Calculus 2 — **Semester 2, 2023**
*MAST90030 Advanced Discrete Mathematics** — **Semester 2, 2023**
*MAST30012 Discrete Mathematics** — **Semester 2, 2022**
*MAST90067 Advanced Methods: Transforms** — **Semester 1, 2022**
*MAST90031 Enumerative Combinatorics** — **Semester 1, 2021**

MAST10007 Linear Algebra	**Semester 1, 2018**
MAST30028 Numerical and Symbolic Mathematics	**Semester 2, 2014**
MAST10005 Calculus 1	**Semester 1, 2014**

The University of Saskatchewan

*MATH 327 Graph Theory**	**Term 2, 2015-2016**
*STAT 241 Probability Theory**	**Term 1, 2015-2016**
*MATH 328 Combinatorics and Enumeration**	**Term 2, 2014-2015**
*STAT 241 Probability Theory**	**Term 2, 2014-2015**

Student & Postdoc Supervision

- Dr. Chris Bradly (Postdoctoral Researcher, 2024-2025)
- Nathan Britt (MSc, completed in 2025)
- Tim Banova (PhD, co-supervised with Prof. Mark Holmes, completed in 2024)
- James Gleeson (MSc, completed in 2024)
- Xin Huang (MPhil, co-supervised with Prof. Mark Holmes, completed in 2023)
- William Briggs (MSc, co-supervising with Prof. Aleks Owczarek, completed in 2022)
- Ruijie Xu (PhD, co-supervised with Prof. Aleks Owczarek and Dr. Richard Brak, completed in 2021)
- Gordon Li (2024), Zhong Zheng (2023) and William Mead (Summer 2017/18) (SCIE30001 undergraduate research projects)

Research & Education Interests

My research interests are in statistical mechanics, particularly lattice models of walks, polygons, animals and trees, and their applications to modelling interacting polymer systems. Problems in these fields are also frequently connected with complex analysis, stochastic processes and algorithms for counting and simulating discrete structures.

I am also interested in the way that changing technology is affecting higher education, and how universities are adapting to the advent of generative AI and mathematical software. The way that students learn is rapidly changing, and the way that we teach and assess mathematics must also evolve.

Grants & Awards

Chief Investigator on Discovery Project (DP) from the Australian Research Council (ARC) **2023-2026**
Title: *Transformative simulation techniques for complex polymer networks*
Budget: $412,000

Discovery Early Career Researcher Award (DECRA) from the ARC **2017–2020**
Title: *Statistical mechanics and topology of polymer systems*
Budget: $337,869

Gavin Brown Prize from the Australian Mathematical Society **2018**
for best paper published in 2008–2016 (see [P7] in the attached Publication List)

Publications

30 articles published in journals in mathematics, physics, and computer science; 6 conference papers; 1 book chapter. More than 30 collaborators. For a full list see my website here.

Conference Organisation

Co-organiser of *GuttmannFest 2025: A conference in celebration of Tony Guttmann's 80th birthday*
June 30 – July 2, 2025
University of Melbourne, Australia

Co-organiser of Special Session on *Topology, geometry and combinatorics of biopolymers*
Joint AustMS / NZMS / AMS Conference 2024
December 9—13, 2024, Auckland NZ

Co-organiser (Proceedings Editor) of the 31st International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2019)
July 1–5, 2019
University of Ljubljana, Ljubljana, Slovenia

Co-organiser (Proceedings Editor) of the 29th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2017)
July 9–13, 2017
Queen Mary University of London, London, UK

Co-organiser of three Contributed Minisymposia on *Combinatorics, topology and statistical mechanics of polymer models*
11th Biennial Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM)
June 1–4, 2015
University of Saskatchewan, Saskatoon, Canada

Selected Invited Presentations

Polymer Modelling and DNA Topology
November 9–14, 2025, Banff International Research Station, Canada

Monte Carlo Algorithms in Statistical Mechanics
June 26–July 7, 2023, MATRIX, Creswick, VIC

A Tour of Combinatorics and Statistical Mechanics in Memory of Richard Brak
February 7, 2022, The University of Melbourne, VIC

Invited Seminar
September 14, 2021, Université Paris 13, France (online)

Transcendence et Combinatoire
January 8, 2021, Université de Versailles-St Quentin, France (online)

The Topology of Nucleic Acids: Research at the Interface of Low-Dimensional Topology, Polymer Physics and Molecular Biology
March 25–29, 2019, Banff International Research Station, Canada

Invited Seminar
June 27, 2018, Colorado State University, Fort Collins, USA

Conference on Means, Methods and Results in the Statistical Mechanics of Polymeric Systems II
June 12–14, 2017, Fields Institute, Toronto, Canada

Monash Workshop on Self-interacting Processes
August 29–31, 2017, Monash University, VIC

Knots and Polymers: Aspects of topological entanglement in DNA, proteins and graph-shaped polymers
August 8–10, 2017, Ochanomizu University, Tokyo, Japan

Outreach, Engagement & Leadership

Member of the *Research and Industry Committee* in the School of Mathematics and Statistics at the University of Melbourne.

Secretary of the Australian and New Zealand Association for Mathematical Physics (ANZAMP, a division of AustMS), 2024-2026.

Member of the Advisory Board for *Journal of Physics A: Mathematical and Theoretical.*

Reviewer for ARC Discovery Project proposals.

Mentor for several students at Elizabeth Blackburn Sciences (University High School) doing Extended Investigation research projects in 2019 and 2020.

Volunteer for *MathsCraft* events, which connect mathematicians with high school students and teachers at events focused on open-ended mathematical problems and thinking like a researcher.

Host of the School Maths Olympics at the 2022 University of Melbourne Open Day.